

BACnet MS/TP, IP and T1L FCU Controller

BACNET PROTOCOL IMPLEMENTATION CONFORMANCE STATEMENT (PICS)

Date: 27-Oct-2023

Vendor Name: Honeywell International Inc.

Product Name: BACnet MSTP, IP and T1L Unitary Controller

Product Model Numbers:

UN-RL1644ES24NM, UN-RL1644ESB24NM, UN-RS0844ES24NM, UN-RS0844ESB24NM, UN-RL1644TS24NM, UN-RL1644TSB24NM, UN-RS0844TS24NM, UN-RS0844TSB24NM, UN-RL1644ES230NM, UN-RL1644ESB230NM, UN-RL1644TS230NM, UN-RL1644TSB230NM, UN-RS0844ES230NM, UN-RS0844ESB230NM, UN-RS0844TS230NM, UN-RS0844TSB230NM, UN-RS0844TSB24NM,

UN-RL1644ES24NMC, UN-RL1644ESB24NMC, UN-RS0844ES24NMC, UN-RS0844ESB24NMC, UN-RL1644TS24NMC, UN-RL1644TSB24NMC, UN-RS0844TS24NMC, UN-RS0844TSB24NMC, UN-RL1644ES230NMC, UN-RL1644ESB230NMC, UN-RL1644TS230NMC, UN-RL1644TSB230NMC, UN-RS0844ES230NMC, UN-RS0844ESB230NMC, UN-RS0844TS230NMC, UN-RS0844TSB230NMC,

UN-RL1644ES24NMD, UN-RL1644ESB24NMD, UN-RS0844ES24NMD, UN-RS0844ESB24NMD, UN-RL1644TS24NMD, UN-RL1644TSB24NMD, UN-RS0844TS24NMD, UN-RS0844TSB24NMD, UN-RL1644ES230NMD, UN-RL1644ESB230NMD, UN-RL1644TS230NMD, UN-RL1644TSB230NMD, UN-RS0844ES230NMD, UN-RS0844ESB230NMD, UN-RS0844TS230NMD, UN-RS0844TSB230NMD

UN-RL1644MS24NM, UN-RL1644MSB24NM, UN-RS0844MS24NM, UN-RS0844MSB24NM, UN-RL1644MS230NM, UN-RS0844MS230NM, UN-RL1644MSB230NM, UN-RS0844MSB230NM, PC-RL1684MS24NM, PC-RL1684MSB24NM,

UN-RL1644MS24NMC, UN-RL1644MSB24NMC, UN-RS0844MS24NMC, UN-RS0844MSB24NMC, UN-RL1644MS230NMC, UN-RS0844MS230NMC, UN-RL1644MSB230NMC, UN-RS0844MSB230NMC,

UN-RL1644MS24NMD, UN-RL1644MSB24NMD, UN-RS0844MS24NMD, UN-RS0844MSB24NMD, UN-RL1644MS230NMD, UN-RS0844MS230NMD, UN-RL1644MSB230NMD, UN-RS0844MSB230NMD

Applications Software Version: BACnet FCU Application

Firmware Revision: 2.0.3.83

BACnet Protocol Revision: 1.16

Product Description:

The BACnet MSTP, IP and T1L Unitary Controllers are a series of unitary controllers that provides flexible, freely programmable, and demand-led control to deliver tangible benefits to reduce energy use and drive new levels of functionality and efficiency in today's buildings. With scalable and freely programmable BACnet MSTP, IP and T1L Unitary Controllers, smart engineering & commissioning tools, and both SYLK and Modbus technology, multiple and flexible configurations can be achieved to address specific applications.

BACnet Standardized Device Profile (Annex L):

- ☐ BACnet Operator Workstation (B-OWS)
- ☐ BACnet Building Controller (B-BC)

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- ☒ BACnet Advanced Application Controller (B-AAC)
- ☐ BACnet Application Specific Controller (B-ASC)
- ☐ BACnet Smart Sensor (B-SS)
- ☐ BACnet Smart Actuator (B-SA)

BACnet Interoperability Building Blocks Supported (Annex K):

Data Sharing:

DS-RP-A	Read Property-A
DS-RP-B	Read Property-B
DS-RPM-B	Read Property Multiple-B
DS-WP-A	Write Property-A
DS-WP-B	Write Property-B
DS-WPM-B	Write Property Multiple-B
DS-COV-A	Change of Value-A
DS-COV-B	Change of Value-B

Alarm and Event Management:

AE-N-I-B	Alarm & Event Notification Internal-B
AE-ACK-B	Alarm & Event Acknowledgement-B
AE-ESUM-B	Alarm & Event Enrollment Summary-B
AE-INFO-B	Alarm & Event Information-B

Scheduling:

SCHED-I-B	Scheduling Internal-B
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Trending

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Device Management

DM-DDB-A	Dynamic Device Binding-A
DM-DDB-B	Dynamic Device Binding-B
DM-DOB-B	Dynamic Object Binding-B
DM-DCC-B	Device Communication Control-B
DM-PT-A	Private Transfer-A
DM-PT-B	Private Transfer-B
DM-TS-B	Time Synchronization-B
DM-UTC-B	Universal Time Synchronization-B
DM-RD-B	Reinitialize Device-B
DM-LM-B	List Manipulation-B

Standard Object Types Supported:

Object Type	Dynamically creatable	Dynamically deletable
Accumulator		
Analog Input		
Analog Output		
Analog Value (no Priority Array)		
Analog Value (with Priority Array)		

Binary Input		
Binary Output		
Binary Value (no Priority Array)		
Binary Value (with Priority Array)		
Calendar		
Device		
File		
Multistate Output		
Multistate Value (no Priority Array)		
Multistate Value (with Priority Array)		
Notification Class		
Schedule		

General Limits

- Object_Name size = 40 characters
- Description size = 132 characters
- Password size for Device Communication Control = 20 characters

Accumulator

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		23
Present_Value	R		
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	writable	
Scale	R		
Units	R		
Max_Pres_Value	R	writable	
Pulse_Rate	O		
High_Limit	O	writable	
Low_Limit	O	writable	
Limit_Monitoring_Interval	O	writable	
Notification_Class	O	writable	range of 0 to 3
Time_Delay	O	writable	range of 0 to 86400s
Limit_Enable	O	writable	
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event_Time_Stamps	O		
Event_Detection_Enable	O		
Property_List	R		

Analog Input

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		0
Present_Value	R		
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 3. no_fault_detected 4. open_loop 5. shorted_loop
Out_Of_Service	R	writable	
Units	R		
COV_Increment	O	writable	
Time_Delay	O	writable	range of 0 to 86400s
Notification_Class	O	writable	range of 0 to 3
High_Limit	O	writable	
Low_Limit	O	writable	
Deadband	O	writable	
Limit_Enable	O	writable	
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event_Time_Stamps	O		
Event_Detection_Enable	O		
Property_List	R		
Terminal Address	P		532 ; adtCharString ; 8 characters.
Sensor Offset	P		533 ; adtReal ;

Analog Output

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		1
Present_Value	W	Writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	Writable	
Units	R		
Priority_Array	R		
Relinquish_Default	R	Writable	
COV_Increment	O	Writable	
Time_Delay	O	Writable	range of 0 to 86400s
Notification_Class	O	Writable	range of 0 to 3
High_Limit	O	writable	
Low_Limit	O	writable	
Deadband	O	writable	
Limit_Enable	O	writable	
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event_Time_Stamps	O		
Event_Detection_Enable	O		
Property_List	R		
Terminal Address	P		532 ; adtCharString ; 8 characters.
PWM Period	P		1017 ; adtReal.
Float Close Runtime	P		1020 ; adtReal.
Float Open Runtime	P		1021 ; adtReal ;
Float Hysteresis	P		1022 ; adtReal ;
Float Powerup Sync	P		1032 ; adtBoolean ;
Float Under Drive Repeat Time	P		1033 ; adtReal ;
Float Sync Operation	P		1037 ; adtEnumerated ;
Float Under Drive Repeat Count	P		1038 ; adtReal ;

Analog Value (no Priority Array)

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		2
Present_Value	R	writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	writable	
Units	R		
COV_Increment	O	writable	
Property_List	R		

Analog Value (with Priority Array)

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		2
Present_Value	W	Writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	Writable	
Units	R		
Priority_Array	O		
Relinquish_Default	O	Writable	
COV_Increment	O	Writable	
Time_Delay	O	Writable	range of 0 to 86400s
Notification_Class	O	Writable	range of 0 to 3
High_Limit	O	writable	
Low_Limit	O	writable	
Deadband	O	writable	
Limit_Enable	O	writable	
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event Time Stamps	O		
Event_Detection_Enable	O		
Property_List	R		

Binary Input

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		3
Present_Value	R		
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	writable	
Polarity	R		
Inactive_Text	O		
Active_Text	O		
Change_Of_State_Time	O		
Change_Of_State_Count	O	writable	
Time_Of_State_Count_Reset	O		
Elapsed_Active_Time	O	writable	
Time_Of_Active_Time_Reset	O		
Time_Delay	O	writable	range of 0 to 86400s
Notification_Class	O	writable	range of 0 to 3
Alarm_Value	O	writable	
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event_Time_Stamps	O		
Event_Detection_Enable	O		
Property_List	R		
Terminal_Address	P		532 ; adtCharString ; 8 characters.

Binary Output

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		4
Present_Value	W	writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	writable	
Polarity	R		
Inative_Text	O		
Active_Text	O		
Change_Of_State_Time	O		
Change_Of_State_Count	O	writable	
Time_Of_State_Count_Reset	O		
Elapsed_Active_Time	O	writable	
Time_Of_Active_Time_Reset	O		
Priority_Array	R		
Relinquish_Default	R	writable	
Time_Delay	O	writable	(range of 0 to 86400s)
Notification_Class	O	writable	range of 0 to 3
Feedback_Value	O		If Feedback_Value is present
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event_Time_Stamps	O		
Event_Detection_Enable	O		
Property_List	R		
Terminal_Address	P		532 ; adtCharString ; 8 characters.

Binary Value (no Priority Array)

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		5
Present_Value	R	writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	writable	
Inactive_Text	O		
Active_Text	O		
Property_List	R		

Binary Value (with Priority Array)

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		4
Present_Value	W	writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	writable	
Inative_Text	O		
Active_Text	O		
Elapsed_Active_Time	O	writable	
Time_Of_Active_Time_Reset	O		
Priority_Array	O		
Relinquish_Default	O	writable	
Time_Delay	O	writable	(range of 0 to 86400s)
Notification_Class	O	writable	range of 0 to 3
Alarm_Value	O		
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event_Time_Stamps	O		
Event_Detection_Enable	O		
Property_List	R		

Calendar

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		6
Present_Value	R		
Description	O		
Date_List	R	writable	
Property_List	R		

Device

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R	writable	
Object_Type	R		8
System_Status	R		
Vendor_Name	R		"Honeywell International Inc. "
Vendor_Identifier	R		17
Model_Name	R		
Firmware_Revision	R		
Application_Software_Version	R		
Location	O	writable	
Description	O	writable	
Protocol_Version	R		1
Protocol_Revision	R		14
Protocol_Services_Supported	R		
Protocol_Object_Types_Supported	R		
Object_List	R		
Max_APDU_Length_Accepted	R		
Segmentation_Supported	R		
Max_Segmenst_Accepted	O		
Local_Time	O	writable	
Local_Date	O	writable	
UTC_Offset	O	writable	
Daylight_Saving_Status	O		
APDU_Segment_Timeout	O		
APDU_Timeout	R		
Number_Of_APDU_Retries	R		
Max_Master	O	writable	
Max_Info_Frames	O	writable	
Device_Address_Binding	R		
Database_Revision	R		
Active_COV_Subscriptions	O		
Last_Restart_Reason	O		
Time_Of_Device_Restart	O		
Serial_Number	O		
Profile_Name	O		
Property_List	R		
Bootloader Version	P		1026 ; adtCharString ; 8 characters.
BLE Configuration Status	P		1031 ; adtUnsigned. 1 = BLE Enabled 0 = BLE Disabled
BLE Connection Status	P		1032 ; adtBoolean. 1 = BLE Device Connected 0 = BLE Device Not Connected
NTP Sync Status	P		1033 ; adtBoolean. 1 = Success 0 = Failed
Application Teach Progress status.	P		1034 ; adtBoolean. 1 = in-progress 0 = not in-progress
Heap memory in use	P		1035 ; adtUnsigned.
DB revision of BACnet FBs	P		1036 ; adtUnsigned.
DB revision of Non-BACnet FBs	P		1037 ; adtUnsigned.

BLE firmware version	P		1272 ; adtCharString ; 8 characters.
Last Firmware Version	P		1056 ; adtCharString ; 8 characters.
Days left for Last Firmware Rollback	P		1057 ; adtUnsigned.
BLE Pass Valid Start Date	P		1058 ; adtDate.
BLE Pass Valid End Date	P		1059 ; adtDate.

File

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		10
File_Type	R		
File_Size	R		
Modification_Date	R		
Archive	W	Writable	
Read_Only	R		
File_Access_Method	R		
Property_List	R		

Multistate Output

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		14
Present_Value	W	writable	
Description	O		
Status_Flags	R		
Event_State	R		
Out_Of_Service	R	writable	
Number_Of_States	R		
State_Text	O		
Priority_Array	R		
Relinquish_Default	R	Writable	
Property_List	R		

Multistate Value (no Priority Array)

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		19
Present_Value	R	writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_OF_Service	R	writable	
Number_Of_States	R		
State_Text	O		
Property_List	R		

Multistate Value (with Priority Array)

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		14
Present_Value	W	writable	
Description	O		
Status_Flags	R		
Event_State	R		
Reliability	O		Possible conditions are: 1. no_fault_detected 2. unreliable_other
Out_Of_Service	R	writable	
Number_Of_States	R		
State_Text	O		
Priority_Array	O		
Relinquish_Default	O	Writable	
Time_Delay	O	writable	(range of 0 to 86400s)
Notification_Class	O	writable	range of 0 to 3
Alarm_Values	O		
Fault_Values	O		
Event_Enable	O	writable	
Acked_Transitions	O		
Notify_Type	O	writable	
Event_Time_Stamps	O		
Property_List	R		

Notification Class

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		15
Description	O		
Notification_Class	R		
Priority	R	writable	
Ack_Required	R	writable	
Recipient_List	R	writable	
Property_List	R		

Schedule

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		17
Present_Value	R	writable	
Description	O		
Effective_Period	R	writable	
Weekly_Schedule	O	writable	
Exception_Schedule	O	writable	
Schedule_Default	R	writable	
List_Of_Object_Property_References	R		
Priority_For_Writing	R	writable	
Status_Flags	R		
Reliability	R		Possible conditions are: 1. no_fault_detected 2. configuration_error
Out_Of_Service	R	writable	
Property_List	R		

Proprietary Objects Supported:

Object Type	Object Type ID
DDC Command Object	512
Output Reference	513
Input Reference	514
Analog Flag	130
Binary Flag	131
Param Obj	132

DDC Command Object

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		512
Present Value	O		
DDC_Command	P		1024

Input Reference

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		514
Description			
Present_Value			Any Boolean Unsigned Real
Reliability			Possible unreliable conditions are: UNRELIABLE_OTHER, COMMUNICATION_FAILURE
Status_Flags			
Units			
Property List	R		

Output Reference

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		513
Description			
Present_Value			Any Boolean Unsigned Real
Priority_For_Writing			
Reliability			Possible unreliable conditions are: UNRELIABLE_OTHER, COMMUNICATION_FAILURE
Status_Flags			
Property List	R		

Analog Flag

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		130
Description			
Present_Value			
Unit			
Property List	R		

Binary Flag

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		131
Description			
Present_Value			
State_Text			
Property List	R		

Parameter Object

Property name	R=Required O=Optional P=Proprietary	Writable	Property ID, Range, Data Type; other remarks
Object_Identifier	R		
Object_Name	R		
Object_Type	R		132
Description			
Present_Value			
Unit			
Default Value	P		1055
Max Value	P		1034
Min Value	P		1035
Property List	R		

Segmentation Capability:

- | | |
|---|----------------------|
| ☒ Segmented requests supported | Window Size <u>4</u> |
| ☒ Segmented responses supported | Window Size <u>4</u> |

Data Link Layer Options:

- ☒ BACnet IP, (Annex J)
- ☐ BACnet IP, (Annex J), Foreign Device
- ☐ ISO 8802-3, Ethernet (Clause 7)
- ☐ ATA 878.1, 2.5 Mb. ARCNET (Clause 8)
- ☐ ATA 878.1, RS-485 ARCNET (Clause 8), baud rate(s) _____
- ☐ MS/TP master (Clause 9), baud rate(s): _____
- ☐ MS/TP slave (Clause 9), baud rate(s): _____
- ☐ Point-To-Point, EIA 232 (Clause 10), baud rate(s): _____
- ☐ Point-To-Point, modem, (Clause 10), baud rate(s): _____
- ☐ LonTalk, (Clause 11), medium: _____
- ☐ Other: _____

Device Address Binding:

Is static device binding supported? (This is currently necessary for two-way communication with MS/TP slaves and certain other devices.) ☐ Yes ☒ No

Networking Options:

- ☐ Router, Clause 6 - List all routing configurations, e.g., ARCNET-Ethernet, Ethernet-MS/TP, etc.
- ☐ Annex H, BACnet Tunneling Router over IP
- ☐ BACnet/IP Broadcast Management Device (BBMD)
Does the BBMD support registrations by Foreign Devices? ☐ Yes ☐ No

Character Sets Supported:

Indicating support for multiple character sets does not imply that they can all be supported simultaneously.

- | | | |
|---|---|-------------------------------------|
| ☒ ISO 10646 (UTF-8) | ☐ IBM™/Microsoft™ DBCS | ☐ ISO 8859-1 |
| ☐ ISO 10646 (UCS-2) | ☐ ISO 10646 (UCS-4) | ☐ JIS X 0208 |

Honeywell

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